FINAL PROJECT

S202322621

Individual Assignment

Phase 1

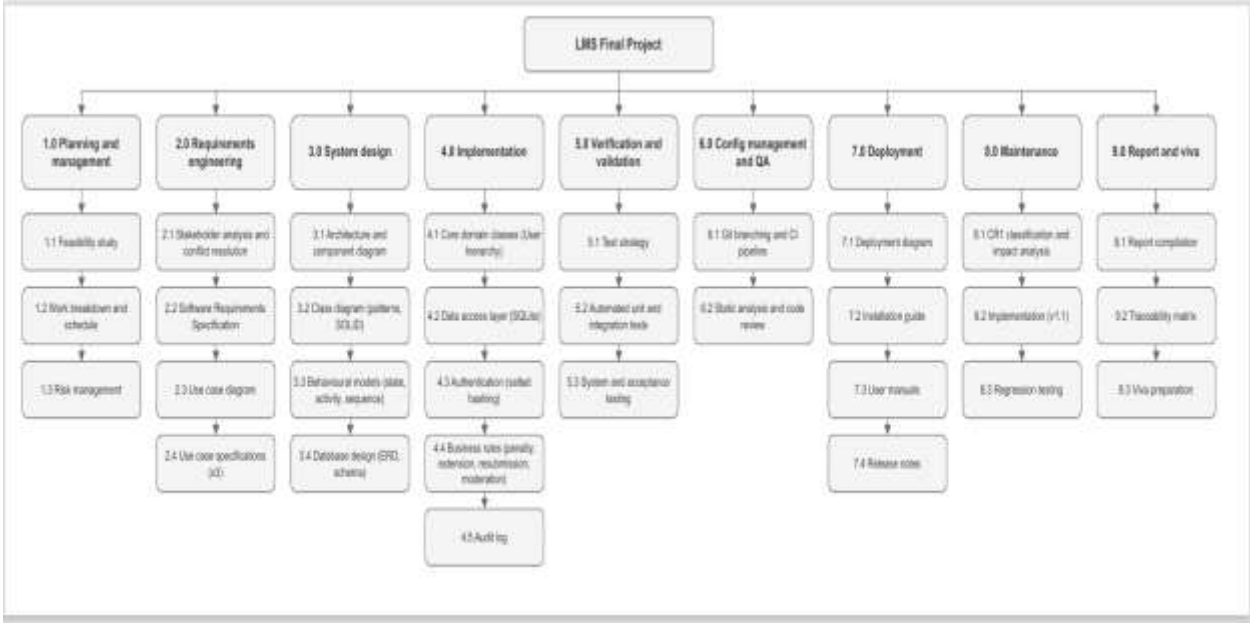
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1. **Feasibilty study**

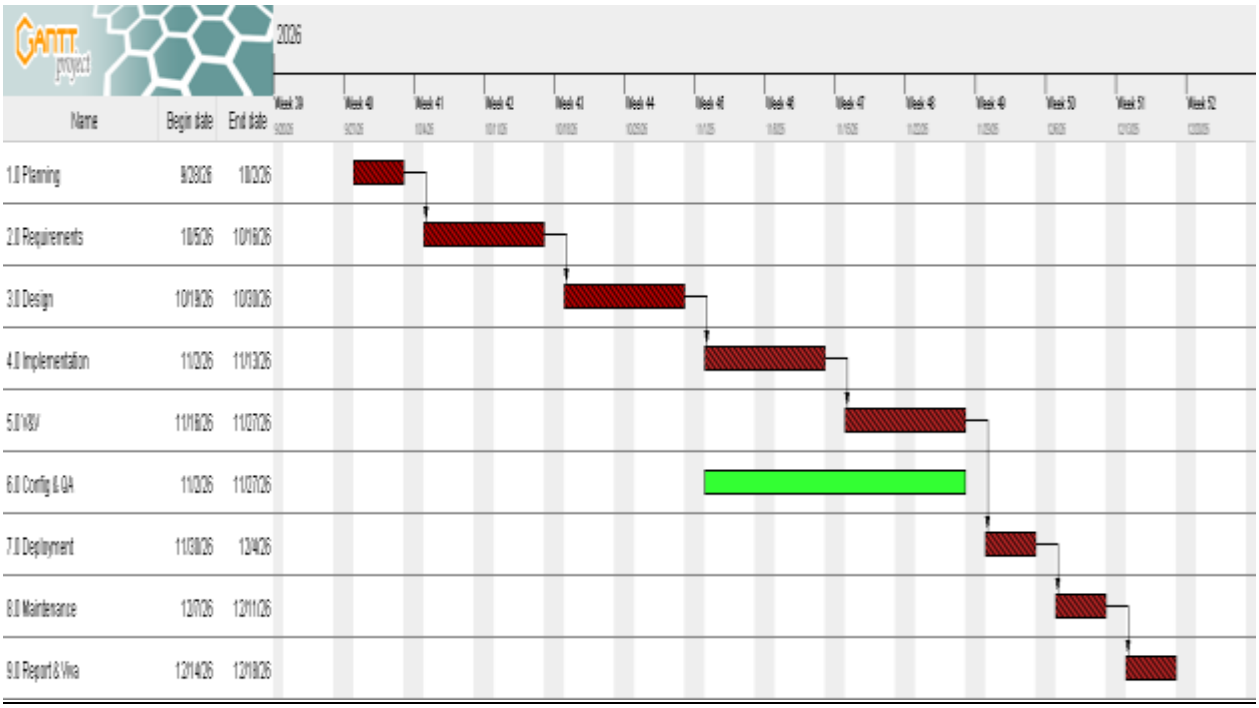
Dimension	Assessment
Technical	Feasible. C++17 supports the object-oriented design needed (an abstract User base class with role-specific derived classes via inheritance and polymorphism). SQLite is embedded and serverless — it requires no installation or network dependency, directly meeting the IT officer's requirement that "our computers are old and the internet is unreliable, so the system must work offline." Alternative considered and rejected: a client–server database (e.g. MySQL/PostgreSQL) — rejected because it introduces a server process and network dependency the offline, single-machine deployment cannot reliably support. Principal risk: correct implementation of design patterns, parameterised SQL queries and salted password hashing — a skill risk, manageable through incremental testing, not a blocking feasibility issue.
Operational	Feasible, conditional on resolving stakeholder conflicts before design begins. The Principal's requirement — "teachers are not IT experts, so the system must be simple" — is treated as a binding, verifiable non-functional requirement, not an aspiration. Key conflict: the Principal requires marks hidden until HoD moderation, while the Lecturer wants marks visible immediately after marking. This is not a technical obstacle but a requirements conflict, resolved explicitly during Requirements Engineering (Section 4.1) before design proceeds — proceeding with it unresolved would make the system operationally unusable for at least one stakeholder group regardless of code quality.
Economic	Feasible. Every tool in the mandated stack is open-source or free for academic use, and the project is delivered by a single student within an existing academic timetable — no direct monetary cost. The relevant cost is opportunity cost of effort, not cash: Requirements and Design (Phases 2–3) together account for 38% of the total grade before implementation begins, so effort must be allocated proportionally rather than rushed toward coding.

Dimension	Assessment
Legal	Feasible, with obligations that shape the design rather than block it. The system stores personal and academic data belonging to minors and young adults (marks, feedback, and an audit trail of who changed a mark, when and why), raising data protection obligations under applicable legislation. Addressed through role-based access control, salted password hashing (never plain-text passwords), and a complete audit log of mark changes — expanded further in the Professional, Legal and Ethical Issues section (Section 11) rather than treated only as a feasibility footnote.
Schedule	Feasible but tight — the single largest feasibility risk identified. Gate 2 (end of Week 3) blocks implementation from starting until design passes review, leaving only Weeks 4–5 for a four-role application with persistent storage, authentication, an audit log and at least two design patterns, before V&V begins in Week 5. Alternative considered and rejected: compressing Requirements and Design into a single combined week to free up implementation time — rejected because it increases the likelihood of design rework discovered only during coding, which costs more to correct than time spent clarifying requirements up front. Mitigation: phase gates treated as hard deadlines, not aspirational targets; CI/testing infrastructure set up from Week 4 rather than deferred to Phase 6.

2. Work Breakdown Structure



3. Gannt Chart



Critical path: 1.0 → 2.0 → 3.0 → 4.0 → 5.0 → 7.0 → 8.0 → 9.0. **6.0 is the only phase off the critical path** — it's explicitly meant to run in parallel (CI pipeline set up during implementation, not after)

5.0 Risk Register

Scoring scheme

Scale	1	2	3	4	5
Likelihood	Rare	Unlikely	Possible	Likely	Almost certain
Impact	Negligible	Minor	Moderate	Major	Severe

ID	Risk	Type	L	I	Exposure	Mitigation	Owner	Review point
R1	Requirements and design phases overrun, compressing the implementation period behind Gate 2.	Project	4	5	20 — High	Treat Gate 2 as a fixed deadline. Timebox each work package in the WBS so Phases 2 and 3 do not expand into the implementation window.	Project Lead	Gate 1, Gate 2
R2	Stakeholder conflicts (mark visibility, late-penalty rule) are resolved incorrectly, causing design rework later.	Technical	3	4	12 — Medium	Resolve each conflict explicitly and in writing during Phase 2, checked against the exact stakeholder statements in the brief, before design begins.	Project Lead	Gate 1
R3	The SQLite schema diverges from the Class Diagram, reducing consistency and traceability.	Technical	3	3	9 — Medium	Design the database and the classes together during Phase 3. Maintain an explicit class-to-table mapping in the design document.	Project Lead	Gate 2
R4	The 70% test-coverage target is missed because testing starts late.	Project	3	3	9 — Medium	Write unit tests alongside implementation rather than deferring them to Phase 5. Track coverage weekly from Week 4.	Project Lead	Gate 3
R5	The salted password-hashing requirement is implemented	Technical	2	5	10 — Medium	Use a recognised library (e.g. libsodium) rather than a custom implementation. Justify the choice explicitly in the report.	Project Lead	Gate 2, Gate 3

	incorrectly or insecurely.							
R6	Illness, hardware failure, or other personal disruption occurs during the seven-week period.	People	2	5	10 — Medium	Commit to the Git repository daily or weekly so work is backed up off-device. No single week should hold irrecoverable, unpushed work.	Project Lead	Ongoing
R7	Scope creep occurs — unrequested features are built at the expense of graded Must requirements.	Project / People	3	3	9 — Medium	Check each proposed feature against the SRS MoSCoW list before building it. Defer anything below Must/Should until core marks are secured.	Project Lead	Gate 2 onward
R8	A design decision cannot be clearly explained at the viva, risking withheld marks for that component.	People	2	4	8 — Medium	Maintain a running decision log throughout development, recording why each significant design choice was made, for review before the viva.	Project Lead	Final weeks

Risk prioritization

R1 is the only risk assessed as High exposure, reflecting its position as the single greatest threat to the project: any overrun in the Requirements or Design phases directly compresses the time available for Implementation, which already carries the least schedule slack of any phase. All other risks are assessed as Medium exposure and are managed through the mitigations listed above. As this is an individual project, the Project Lead is the sole owner of every risk; accountability for monitoring and mitigation therefore rests with a single person rather than being distributed across a team, which is reflected consistently in the Owner column.